

Response to RFI 80TECH26RFI0020

Strategic Partnerships for NASA Data Center Infrastructure

Field	Response
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Platform	LumenCore evidence-first validation platform
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Executive Summary

Robert Ashworth proposes LumenCore as a bounded validation and evidence workstream within a larger NASA data center modernization partnership. LumenCore is not presenting itself as a data center construction prime, cloud service provider, cybersecurity authorization provider, or current NASA operator. Its role is narrower: help NASA and an authorized prime or technical partner compare modernization concepts against locked baselines before operational commitment.

The proposed approach creates a reproducible decision record for each candidate concept. It identifies the source data used, the incumbent or reference baseline, the metric and acceptance rule selected before evaluation, the candidate workflow tested, the observed result, the uncertainty and failures, and the human decision to advance, revise, or stop. This can support phased modernization, hybrid workload planning, resilience exercises, energy and water measurement, vendor-claim review, and pilot acceptance without granting an automated system operational authority.

The recommended entry point is a limited proof-to-pilot work package using one non-sensitive historical operations window, one NASA-approved baseline, and one modernization question. The result would be an inspectable evidence bundle, not a production deployment or unsupported savings claim.

1. Partnership Models and Value Creation

LumenCore is best positioned as a small-business validation partner or subcontracted evidence workstream inside a team led by an organization with demonstrated data center engineering, facilities, cloud, cybersecurity, and mission-operations capability.

Three phased structures could create mutual value:

- **Validation work package within a FAR procurement:** the prime performs engineering and delivery; LumenCore freezes the evaluation protocol, runs authorized candidate-versus-baseline comparisons, and produces reviewer receipts.

- **Pre-procurement proof exercise:** NASA supplies a bounded non-sensitive question and approved historical data; LumenCore returns a reproducible result and limitations register that NASA may use in market research.
- **Research collaboration under an authority selected by NASA:** NASA and partners study measurement, routing, or evidence methods without implying a procurement commitment.

NASA would retain operational authority, data ownership, security decisions, and acceptance authority. The partner team would retain responsibility for engineering, facilities, cloud, and cybersecurity obligations. LumenCore would be responsible only for the agreed validation protocol, execution receipts, and evidence package.

Risk should be released in gates: source readiness, baseline and metric lock, offline replay, controlled pilot, independent review, then an operational decision. A failed or inconclusive result should stop or revise the candidate without forcing a larger commitment.

2. Future Data Center Concepts

A next-generation NASA data center environment is likely to combine on-premises mission systems, approved cloud services, edge or distributed resources, high-density compute, and facilities constraints. The useful architectural question is therefore not whether one platform replaces all others, but which workload and infrastructure pattern is appropriate under which mission, security, latency, resilience, energy, and cost constraints.

LumenCore can support this decision by representing each option as a candidate policy or workflow and replaying it against the same frozen operating window. Candidate concepts may include workload placement, failover policy, storage tiering, cooling or energy heuristics, maintenance sequencing, and capacity allocation. Results would preserve regime-specific performance rather than collapsing every condition into one score.

The platform is designed to complement, not replace, engineering models, digital twins, facilities telemetry, cloud observability, and cybersecurity tools supplied by NASA or qualified partners.

3. Modernization of Legacy Facilities

Legacy modernization should proceed through bounded increments that separate measurement from control:

Gate	Activity	Decision evidence
Baseline	Inventory approved telemetry, constraints, incidents, and current operating policy.	Source quality ledger and incumbent metric card.
Candidate design	Define one retrofit, modular, software-defined, co-location, or hybrid option.	Candidate specification and predeclared acceptance rule.
Offline replay	Evaluate the candidate on the same historical operating window.	Candidate-versus-baseline results, uncertainty, failures, and provenance.
Controlled pilot	Test within a limited, reversible, human-supervised boundary.	Pilot receipt, rollback record, and completion criteria.
Scale decision	Advance, revise, or stop.	Signed decision record and unresolved-risk register.

This structure helps preserve useful existing assets, identify where data is too weak to support a decision, and avoid treating a polished demonstration as production evidence.

4. Reliability, Resilience, and Security

Reliability and resilience evaluation should include normal load, peak mission demand, component loss, network degradation, telemetry loss, recovery, and rollback scenarios selected by NASA and its engineering partners. Metrics should include availability or continuity measures appropriate to the system, recovery time, failed-action count, constraint violations, and uncertainty.

Automation should remain bounded by explicit human authority. A candidate may recommend or simulate an action, but a NASA-authorized operator should approve any operational change. Every recommendation should preserve its source, model or rule version, confidence, constraint status, and final human disposition.

LumenCore does not currently claim FedRAMP authorization, FISMA authorization, CMMC assessment authority, classified-system access, CUI handling approval, or NASA cybersecurity validation. Any work involving sensitive systems would run only inside an architecture and data boundary approved by NASA and delivered with appropriately authorized partners. Public proof artifacts would contain no sensitive NASA data.

In a mixed government and commercial environment, isolation must be enforced by the authorized infrastructure and cybersecurity architecture. LumenCore's contribution would be to preserve test evidence showing which isolation, resilience, and recovery assumptions were evaluated and under what configuration.

5. Operational and Service Models

For mission-critical mixed-use infrastructure, the operating model should distinguish routine periods, planned high-priority mission windows, contingencies, and recovery. Each period can have its own constraints, service objectives, escalation paths, and allowed automation level.

An evidence-first service layer would support:

- configuration and source-version receipts for each evaluation;
- mission-window-specific baselines and acceptance rules;
- change, exception, and rollback records;
- human approval for any action that affects operations;
- an append-only limitations and negative-results register; and
- exportable machine-readable and human-readable review packages.

This model supports accountability across NASA, a prime contractor, facilities operators, cloud providers, and specialist small businesses without asking one dashboard to become the system of record for every domain.

6. Scalability and Future Demand

The proposed validation layer scales by adding adapters and candidate policies around a stable evidence contract: source, baseline, candidate, metric, result, uncertainty, constraint, hash, and

reviewer decision. The same contract can evaluate new workload classes or infrastructure options while retaining prior results for comparison.

To reduce vendor lock-in, source and result records should use documented, exportable schemas; retain raw source references; identify proprietary dependencies; and permit the core evaluation bundle to run in more than one approved environment where feasible. Portability should be tested with identical frozen inputs and compared by result hash, rather than assumed from architecture diagrams.

Capacity planning should report regime behavior and failure thresholds. A candidate that performs well at current load but fails at a future density, latency, water, power, or resilience constraint should be marked conditional, not promoted as a universal winner.

7. Sustainability and Efficiency

Energy, water, and cost decisions should be tied to NASA-approved telemetry and accounting boundaries. Candidate metrics may include PUE, WUE, energy per useful workload, cooling-system constraint violations, peak-demand contribution, capacity utilization, or another measure selected by the facility owner. Proxy metrics should be labeled as proxies and never converted into savings without an approved economic model.

For each modernization option, LumenCore would preserve:

- the measurement interval and facility or workload boundary;
- data completeness and sensor-quality checks;
- the incumbent baseline and weather or load regime;
- the candidate policy and all constraints;
- effect size, uncertainty, and failure cases; and
- the approved method, if any, for converting a technical delta into dollars.

No energy, water, uptime, or cost improvement is claimed in this response.

Evidence and Differentiators

LumenCore's differentiator is not a claim that one model always wins. It is an evidence process that can retain losses as clearly as wins.

An exploratory broad benchmark completed 2,172 frozen public and market series across naive, linear, harmonic, neural, tree, and classical model families. Harmonic models led on 304 series, while other families led more often overall. The correct conclusion was that harmonic models may be useful routed specialists, not universal replacements.

A later preregistered benchmark used 14,704 official U.S. Energy Information Administration panel rows. The protocol and candidate-selection rules were committed before the final holdout. On that holdout, the official EIA day-ahead forecast and an autoregressive ridge baseline both beat every tested wave-geometry candidate. LumenCore retained the negative result and blocked the earlier synthetic Kuramoto result from becoming an unsupported grid-performance claim.

A second preregistered test kept the EIA forecast as the incumbent and asked whether fixed residual learners could correct it. XGBoost residual correction was selected on development data only. On the untouched holdout, its mean seasonal MASE was 0.212, compared with 0.579 for the EIA forecast, 0.264 for direct XGBoost, and 0.236 for direct LightGBM. It still failed LumenCore's full champion gate

because its regression against autoregressive ridge on one balancing authority exceeded the predeclared guardrail. The result therefore supports conditional residual routing research, not a universal-winner claim.

These are internal technical analogs, not NASA performance, field validation, realized savings, or agency endorsement. They demonstrate the intended review behavior: freeze the question, compare against strong baselines, report all outcomes, and stop unsupported promotion.

Proposed Limited Pilot

NASA or an authorized partner would select one non-sensitive historical operating window, one incumbent baseline, one candidate modernization workflow, and one primary metric. A compact pilot could proceed as follows:

- Confirm data rights, security boundary, operational owner, and decision question.
- Inventory and hash approved source records; report quality and missingness.
- Lock the baseline, candidate, metric, exclusions, tie rule, and completion criteria.
- Run the candidate and baseline on the same frozen window.
- Report effect size, uncertainty, failures, constraint violations, and regime behavior.
- Deliver a reviewer packet and a recommendation to advance, revise, or stop.

Expected outputs would include a source and quality ledger, protocol, baseline card, candidate replay report, negative-results register, machine-readable manifest, artifact-chain hash, and a human-readable decision brief.

Capability Boundaries

- No pricing is included.
- No NASA deployment, access, validation, endorsement, or past performance is claimed.
- No autonomous operational control is proposed.
- No FedRAMP, FISMA, CMMC, CUI, classified, safety, or cybersecurity authorization is claimed.
- No realized cost, energy, water, reliability, or uptime improvement is claimed.
- Detailed facilities engineering, cloud delivery, and security authorization would require qualified partners.

Closing

Robert Ashworth welcomes consideration of LumenCore as a focused small-business validation workstream within a qualified NASA data center modernization team. The practical first step is deliberately small: one approved question, one frozen source window, one incumbent baseline, one candidate, and one decision record that NASA can inspect and accept or reject before broader commitment.